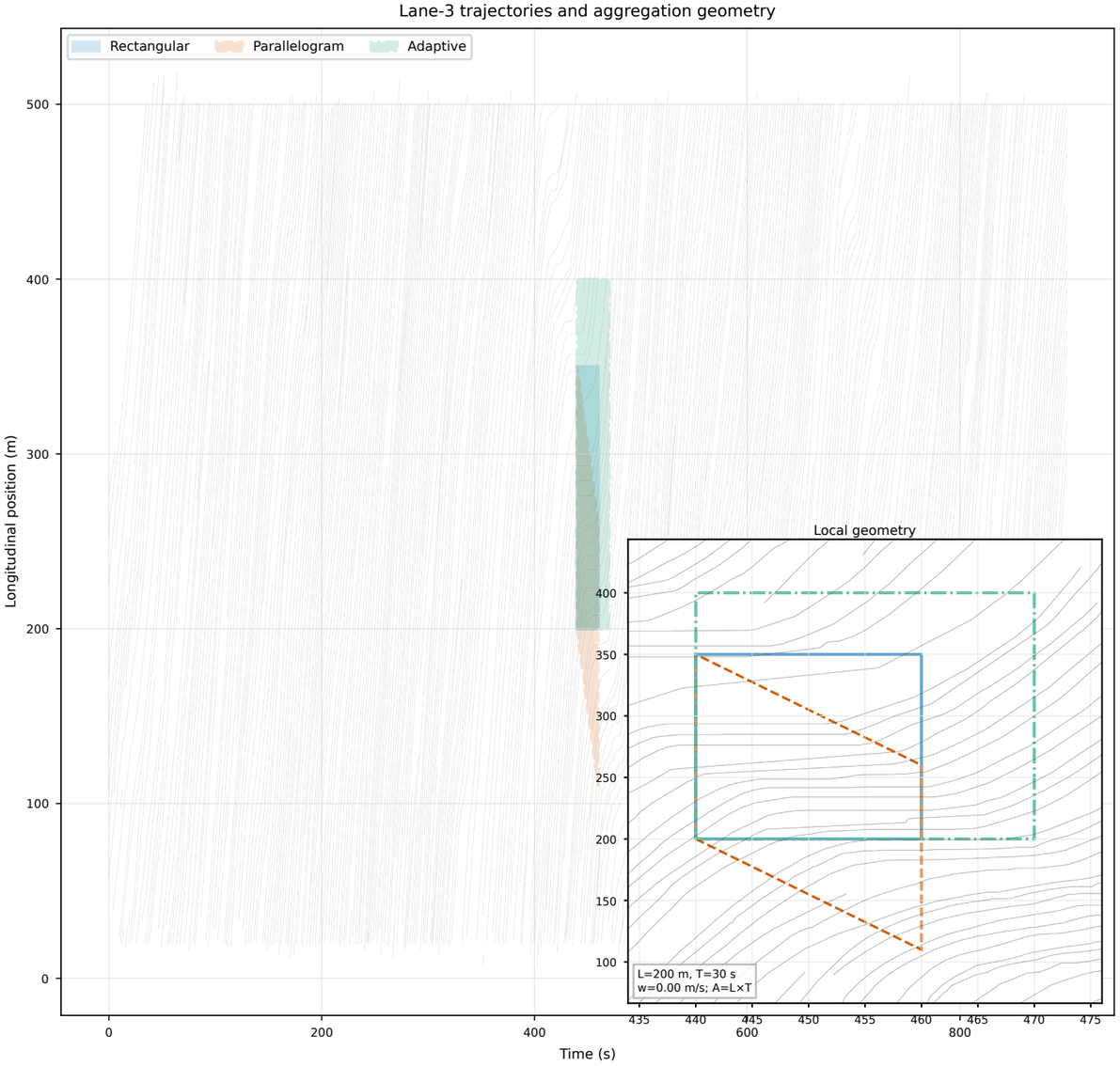
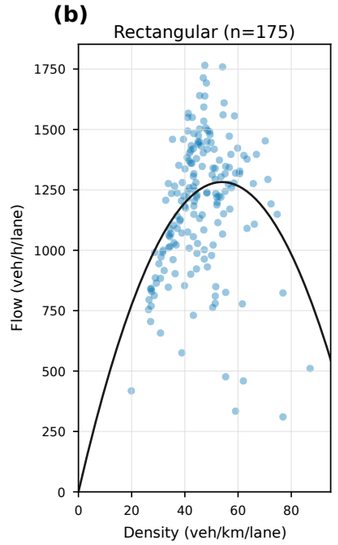
Empirical Fundamental Diagrams from NGSIM I-80 Trajectories

Objective: compare exact Edie states from rectangular, wave-aligned, and exploratory adaptive windows on a common lane/time-space domain.

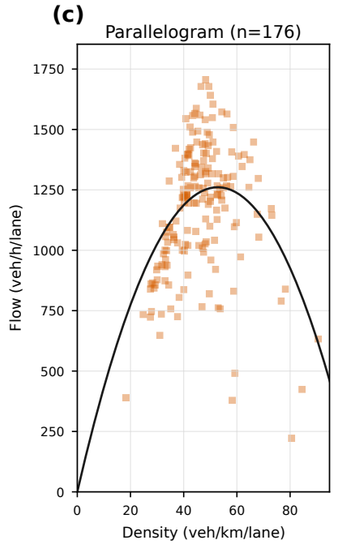
(a)



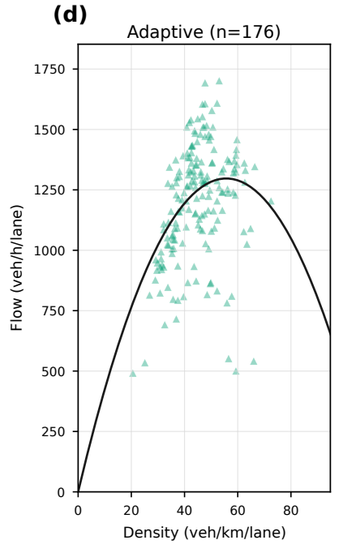
(b)



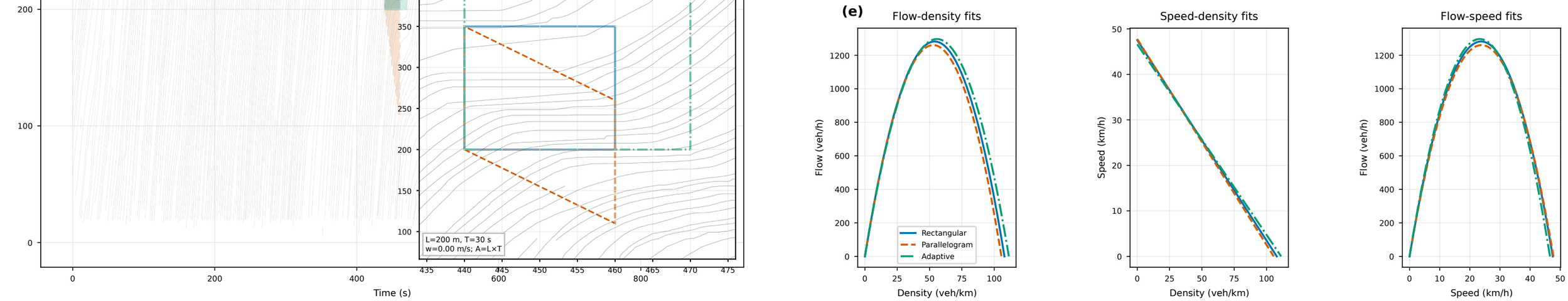
(c)



(d)



(e)



(f)

Quantitative comparison (held-out final 30% time blocks)

Method	Samples	MAE (veh/h)	RMSE (veh/h)	Capacity (veh/h)	Stability CV
Rectangular	175	197	245	1,282	13.1%
Parallelogram	176	186	238	1,261	43.2%
Adaptive	176	175	234	1,297	23.3%

(g)

Key findings

Wave-aligned windows used an empirical w=-4.50 m/s; the adaptive method selected size and orientation without discarding acceleration/deceleration states.
Adaptive had the lowest held-out flow RMSE (234 veh/h) under the common time-block split.
Valid windows (Rectangular: 175/176, Parallelogram: 176/176, Adaptive: 176/176). Lower scatter or RMSE alone does not prove more accurate state extraction; results are limited to one lane and one 15-min period.

Lane 3; SI units; exact segment-boundary interpolation. Parallelogram geometry is a simplified implementation inspired by He & Wu (2025), not a full reproduction. Greenshields fits are reference models, not extraction accuracy.